QUESTION 5:

1) We could either use a linear regulator or DC-DC Buck Converter.
   Linear regulator: An internal circuit reads the voltage that enters the ESP32 if it goes more than 3.3V it constricts the current flow dissipating the rest of the Battery Power as heat. It is very inefficient as the battery unloads its max power and most of it is dissipated as heat. It also heats up very quickly to high temperatures. The resistance of a linear regulator that uses a BJT is regulated by decreasing or increasing the base current.
   The main components of a linear regulator are the zener diode, op-amp and BJT. The current flows from the battery via a safety resistor to the collector plate, whereas a small current trickles to the zener diode, the diode sets up a potential of 3.3/2 V at the op-amps one terminal. The current passes through the collector to emitter and is fed back into another terminal of the op amp, this forms a feedback system. Essentially the op amp compares the two terminals and controls the base current of the transistor suitably so as to ensure both the output voltage is 3.3V

   DC-DC Buck Converter: Essentially uses the principle of PWM. Just like how PWM is used to control fan speed or LED light intensity, it is also used to control the effective power and thus effective voltage that flows into ESP32. The battery connection is switched on and off rapidly creating a Duty cycle of (3.3/12)*100=27.5%. This is done without dissipating huge amounts of the battery's power.
   When the battery switch is closed current flows into an inductor and is trapped as magnetic energy. A choked current goes to the capacitor which smoothens out and supplies it to the main ESP board.Based on the duty cycle the battery is then switched off, the inductor doesn't like sudden changes and hence uses its magnetic energy and supplies current and also at this phase the diode is forward biased and to complete the circuit, the diode allows current from ground line to flow. Since it stores and reuses energy it has a higher efficiency.
2) Yes the TB6612FNG can manage up to a maximum voltage of 15V according to the datasheets. https://cdn.sparkfun.com/datasheets/Robotics/TB6612FNG.pdf . A fully charged Li battery will give a max output of 12.6V which sits comfortably beneath the threshold. So Yes the driver will survive a fully-charged battery.
3) I picked the pins GPIO 21 and 22 for I2C communication. This is precoded in the hardware of the ESP to be used as communication pins and hence by nature it is faster and more efficient to use this to communicate with the sensor. GPIO 34 and 35 are purely input pins they cannot be used for output and since the motor encoder would just send data and is not required to take any action, these pins are more suitable. The driver input pins are connected to GPIO 25,26 and 27 as all of these pins can be used to output PWM signals which is required to control the motor speed
4) The driver has 2 power supplies one (12V) goes directly to the motors based on the input signals from ESP and the other (3.3V) that is used to power the logical analyzer of the motor driver to process the ESP32 signals. The 12V flows into the motor then back to the driver where it is finally passed to the common ground rail. We could use

MOSFET or BJT to prevent damage to components because of reversing battery polarity. The gate is connected to the ground line and the source to the power rail. When the battery is connected appropriately then the difference between gate voltage and source voltage would be sufficient to turn the channel on and let current flow, whereas when the polarity is reversed the potential difference between gate and source is also reversed and thus the drain is blocked and no current flows into the circuit.